VISHWANI SINGH

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Education

Northeastern University, Boston, USA, *MS, Artificial Intelligence (4.0/4.0 GPA)* **Jan 2025 – Expected Dec 2026**
Institute of Engineering and Technology, IN, *B.Tech, Information Technology (8.52/10 GPA)* **2017 – 2021**

Experience

Massachusetts General Hospital **Jan 2026 – Present**
Applied AI Scientist Co-op, OASIIS lab *Boston, MA*

- Lead development of an IMU-based foundation model designed specifically for clinical populations to close the gap where general-population models underperform on patients.
- Designed and built an evaluation benchmark assessing four pretrained foundation models — SSLWearables (UK Biobank wrist-accel SSL), UniMTS (skeleton-aware CLIP-style IMU), MOMENT (general time-series Transformer), and ImageBind (egocentric IMU) - on a phase-1 patient-vs-control classification task using in-house clinical IMU data, evaluating each model under frozen linear probing and full encoder fine-tuning across full-shot and few-shot regimes to test whether general-population pretraining transfers to a clinical cohort.

Graduate Research Assistant (Supervisors: Prof. Aarti S., NEU & Dr. Christopher Bono, MGH) **Mar 2025 – Dec 2025**

- Developed an HMM-based biomechanical analysis pipeline on XR movement sequences and a laterality-based validation framework to identify relevant movement signatures, narrowing 24 task-paradigm combinations to optimal subsets. [Singh et al., HMM for Biomechanical Phenotyping and Classification of Shoulder Pathology in XR, 2026, Communicated]
- Established deep learning baselines (1d CNN, LSTM/GRU and Vanilla Transformer) on selected subsets using LOSO-CV across 60 subjects, achieving 85–91% balanced accuracy from 6-DoF IMU sequences. [Singh et al., Machine Learning Classification of Shoulder Pathology Within a Clinically Informed Extended Reality-Based Functional Assessment Framework: Proof-of-Concept Study, 2026, JMIR XR and Spatial Computing, Under Review]

Indian Institute of Technology, Delhi **Sep 2023 – Jul 2024**
Senior Research Fellow (Supervisor: Prof. Tapan K. Gandhi, IIT Delhi) *Delhi, IN*

- Developed graph-theoretic EEG functional connectivity algorithms to quantify tDCS treatment effects and identify neural biomarkers in MDD patients. [Technical Report, Singh et al., 2024]

GeeksForGeeks **Sep 2021 – Aug 2023**
Software Engineer II *Noida, IN*

- Built and optimized scalable backend infrastructure for ML-driven search and ranking pipelines, integrating AWS OpenSearch and Redis to support high-traffic content discovery workflows serving $\sim 1M$ assets.

Technical Skills

- **Programming & Systems**: Python, SQL, Django, DRF, React.js, REST APIs, Linux, GPU computing (CUDA, Torch), HPC clusters, Git, Github actions, Docker, Amazon Web Services (S3, OpenSearch, EC2), Redis, ElasticSearch
- **Machine Learning & AI**: Deep Learning, Self-Supervised Learning, Representation Learning, Transfer Learning, Time Series Analysis, Signal Processing (EEG/IMU), Transformers, Mixture-of-experts, LSTM/GRU, CNN, HMM, Graph-Theory, RAG, LLMs, Hugging Face, scikit-learn, Pandas, numpy, matplotlib, Weights & Biases
- **Clinical & Research**: Digital Biomarker Discovery, Computational Neuroscience, Clinical AI, Translational AI, Biomechanical Analysis, XR-based Movement Assessment, LOSO Cross-Validation, Ablation Studies, Permutation Testing, Hyperparameter Tuning

Projects

Subject-Invariant EEG Decoding – Designed a subject-invariant EEG decoding model for the NeurIPS 2025 EEG Challenge to predict reaction time across subjects for generalizable BCIs. Integrated spatial filters, multi-temporal convolutional kernels, a Mixture-of-Experts architecture, and custom load-balancing losses. Achieved RMSE of 0.3609, outperforming the EEGSimpleConv baseline of 0.3914.

Transformer-Based Soccer Action Spotting – Developed a Transformer-based temporal action spotting pipeline with positional encoding and Multi-Head Attention pooling (8 heads) on SoccerNet v2 (471 matches, 716+ hours) using pre-extracted ResNet embeddings (512-dim, 2 fps). Achieved 34.7% validation mAP, nearly 2× the AVG pooling baseline (17.8%).

Human-Centered AI Dashboard for Clinical XR Systems – Led user research to evaluate an AI-assisted XR dashboard for explainable clinical decision-making. Built a Gemini-powered RAG chatbot grounded in clinical guidelines and motion assessment protocols for clinician and XR expert queries. Proposed a 3-stage XAI framework - Glance, Scan, Investigate - for progressive clinician engagement with AI explanations.